MIP_TECHNICAL_DECISIONS.md

Mainframe Intelligence Platform

Architecture Decision Records (ADR)

Version: 1.0

Purpose

This document records key architectural decisions made during the design and implementation of MIP.

Goals:

- Preserve architectural reasoning
- Avoid repeated debates
- Enable onboarding
- Provide leadership transparency
- Support long-term evolution

ADR-001

Graph First Architecture

Status:

Accepted

Context

Most modernization initiatives use:

Source Code ↓ LLM ↓ Summary

This approach creates:

- Hallucinations
- Inconsistent answers
- Lack of traceability
- Weak impact analysis

Decision

MIP will use:

Source Code ↓ Metadata ↓ Knowledge Graph ↓ Reasoning ↓ AI

Rationale

Benefits:

- Deterministic
- Explainable
- Traceable
- Scalable

AI becomes a consumer of knowledge rather than the source of truth.

Consequences

Pros:

- Higher confidence
- Better impact analysis
- Better lineage

Cons:

- Additional implementation effort

ADR-002

Metadata Repository + Knowledge Graph

Status:

Accepted

Context

Two common approaches:

Option A

Everything in Graph

Option B

Everything in Relational Database

Decision

Use both.

Metadata:

PostgreSQL

Relationships:

Neo4j

Rationale

PostgreSQL:

- Structured storage
- Reporting
- Transactional consistency

Neo4j:

- Relationship traversal
- Dependency analysis
- Impact analysis

Each technology solves a different problem.

ADR-003

Python as Primary Platform Language

Status:

Accepted

Context

Candidate Languages:

- Java

- Python
- Go

Decision

Python 3.13

Rationale

Benefits:

- Fast development
- Excellent parsing ecosystem
- Strong AI ecosystem
- Strong graph ecosystem
- Large talent pool

Risks

Potential CPU bottlenecks.

Mitigation:

- Horizontal scaling
- Async processing
- Future Rust extensions if required

ADR-004

FastAPI for Service Layer

Status:

Accepted

Decision

FastAPI

Rationale

Benefits:

- High performance
- OpenAPI support
- Strong typing
- Pydantic integration

ADR-005

Pydantic as Domain Contract

Status:

Accepted

Decision

All metadata models use Pydantic.

Rationale

Benefits:

- Validation
- Serialization
- Type safety
- API consistency

ADR-006

Neo4j as Initial Graph Store

Status:

Accepted

Context

Candidates:

- Neo4j
- Neptune
- Cosmos DB Graph
- JanusGraph

Decision

Neo4j

Rationale

Benefits:

- Mature ecosystem
- Cypher language
- Visualization support
- Strong community

Future Review

Evaluate Neptune for large enterprise deployments.

ADR-007

Plugin-Based Parser Architecture

Status:

Accepted

Decision

Every parser implements a common contract.

Examples:

- COBOL Parser
- JCL Parser
- DB2 Parser

- VSAM Parser

Rationale

Benefits:

- Extensibility
- Isolation
- Independent testing

ADR-008

Monorepo Strategy

Status:

Accepted

Decision

Single repository.

Rationale

Benefits:

- Shared models
- Simpler refactoring
- Easier onboarding
- Consistent CI/CD

Future Review

Evaluate multi-repo after platform maturity.

ADR-009

Incremental Discovery

Status:

Accepted

Decision

Process changes incrementally.

Avoid full repository rescans.

Benefits

- Faster updates
- Lower costs
- Better scalability

ADR-010

AI Last Principle

Status:

Accepted

Decision

AI is introduced only after:

1. Metadata
2. Graph
3. Reasoning

are established.

Rationale

Most modernization projects fail because AI is introduced before enterprise knowledge is captured.

ADR-011

Business Capability-Centric Modeling

Status:

Accepted

Decision

Model business capabilities as first-class entities.

Examples:

- Payments
- Authorization
- Rewards
- Fraud

Rationale

Modernization targets business capabilities, not programs.

ADR-012

Event-Driven Internal Architecture

Status:

Proposed

Decision

Use Kafka for asynchronous processing.

Benefits

- Scalability
- Decoupling
- Continuous ingestion

Initial MVP

May start with synchronous workflows.

ADR-013

LangGraph for Agent Orchestration

Status:

Proposed

Decision

Use LangGraph for future reasoning agents.

Benefits

- Workflow orchestration
- State management
- Human-in-the-loop

MVP

Not required.

ADR-014

Enterprise Observability

Status:

Accepted

Decision

Every service must emit:

- Metrics
- Logs
- Traces

Stack

- OpenTelemetry

- Prometheus
- Grafana

ADR-015

Vendor-Neutral Modernization Strategy

Status:

Accepted

Decision

MIP must remain independent of:

- Cloud provider
- LLM provider
- Mainframe vendor

Benefits

- Portability
- Lower lock-in
- Broader adoption

Guiding Rule

Whenever a new architectural decision is made:

Create a new ADR.

Do not rely on tribal knowledge.

The ADR repository becomes the institutional memory of MIP.